EEE-301
Data and Telecommunication

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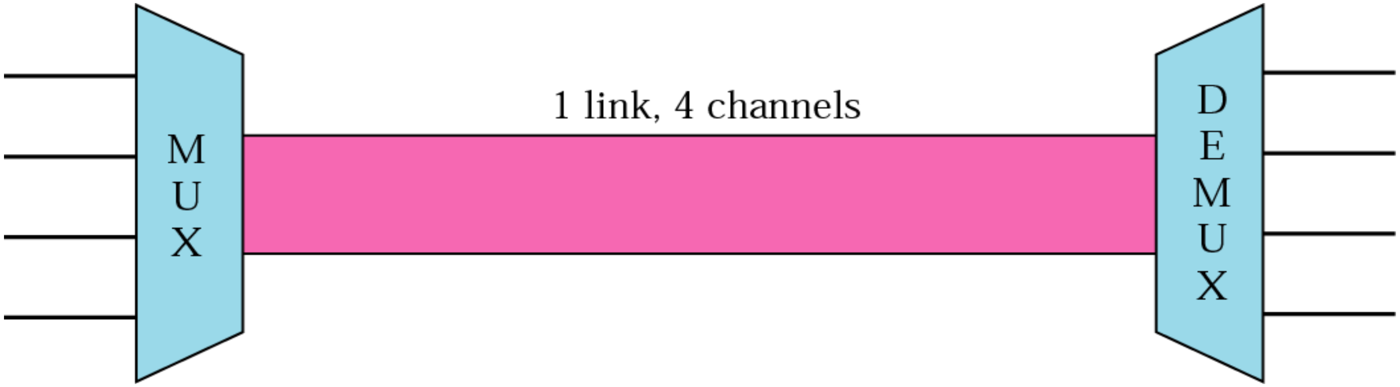
Chapter-6

Bandwidth Utilization: Multiplexing and Spreading

Multiplexing

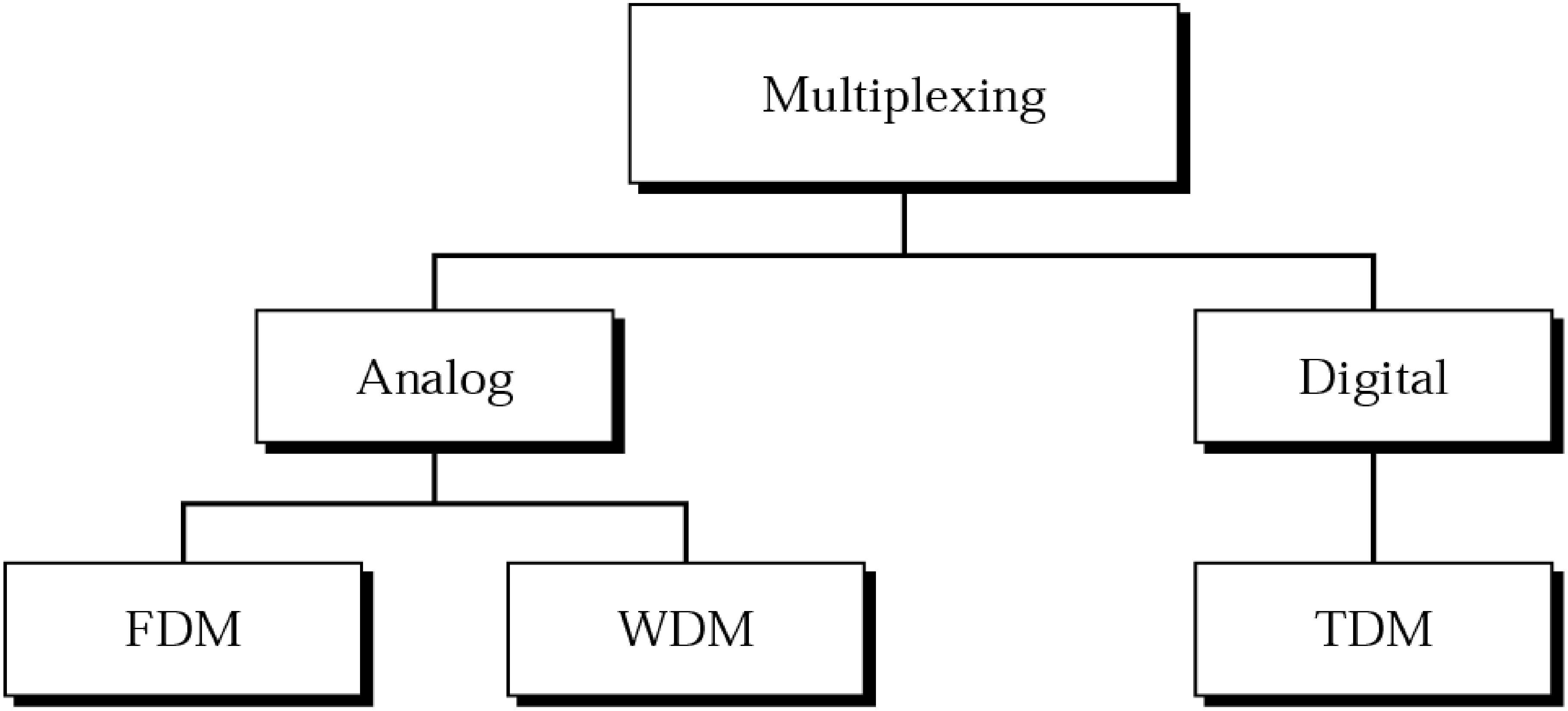
- **Multiplexing** (sometimes contracted to **muxing**) is a method by which multiple analog or digital signals are combined into one signal over a shared medium.
- For example, in telecommunications, several telephone calls may be carried using one wire.
- Multiplexing originated in telegraphy in the 1870s, and is now widely applied in communications.

Multiplexing



- Set of techniques that allow the simultaneous transmission of multiple signals over a single link.

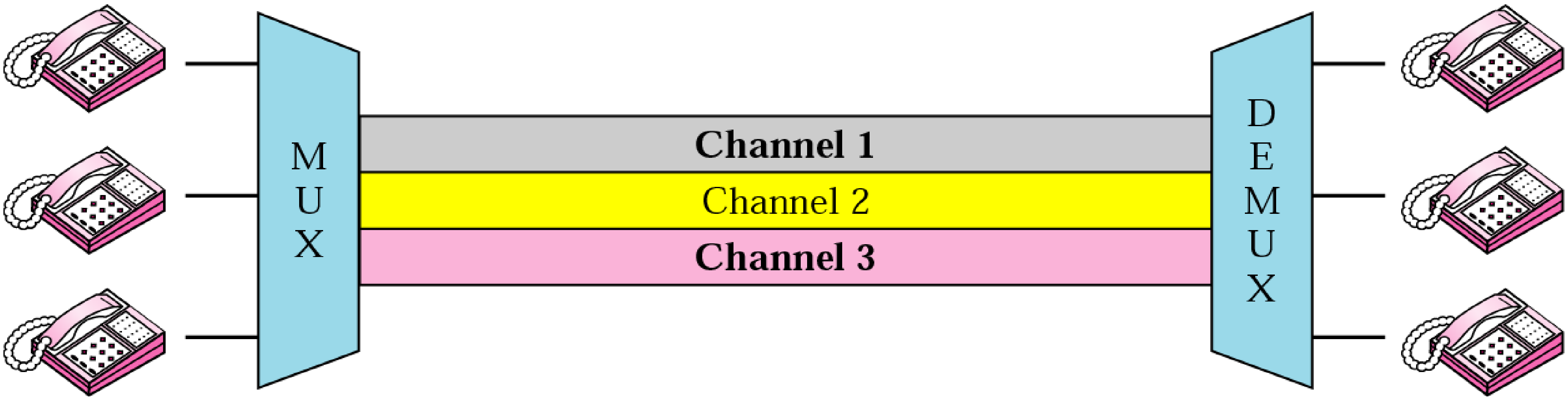
Categories of Multiplexing



Frequency-division multiplexing (FDM)

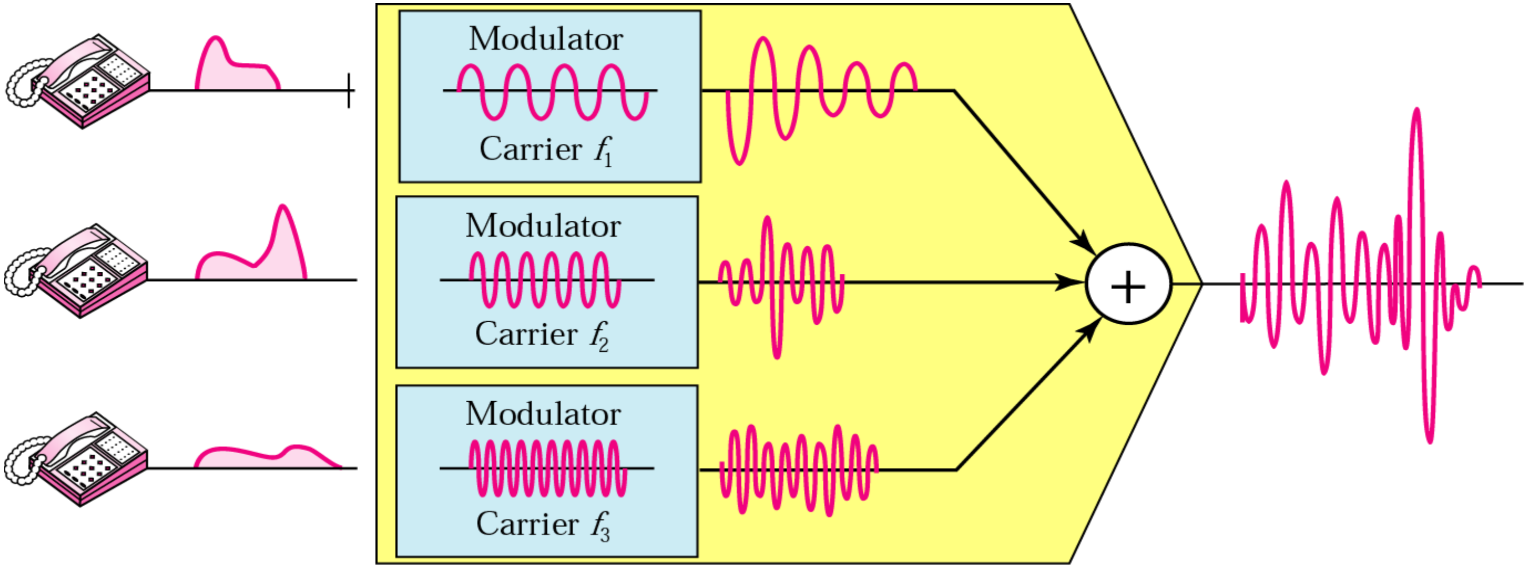
Frequency-division multiplexing (**FDM**) is a technique by which the total bandwidth available in a communication medium is divided into a series of non-overlapping frequency bands, each of which is used to carry a separate signal. This allows a single transmission medium such as a cable or optical fiber to be shared by multiple independent signals. Another use is to carry separate serial bits or segments of a higher rate signal in parallel.

Frequency Division Multiplexing (FDM)



- Used when link bandwidth (hz) > the combined bandwidth of the signals to be transmitted.

FDM: Multiplexer



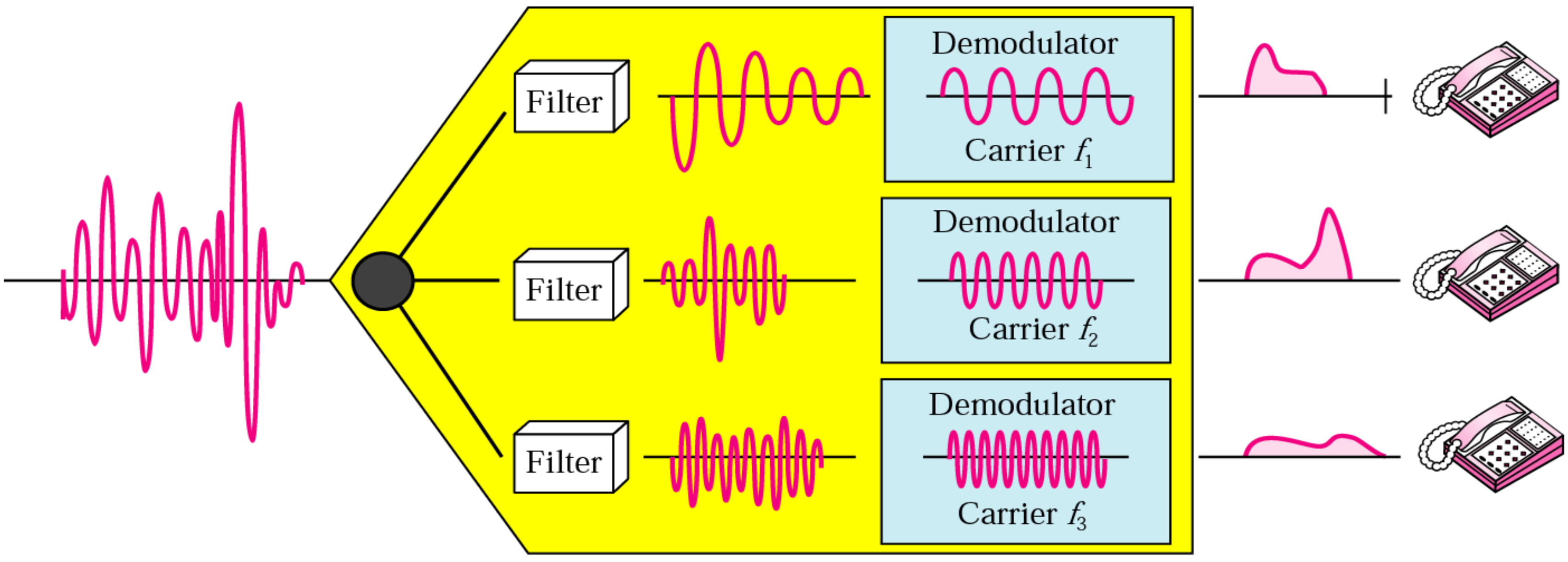
<u>Modulation step</u>:

1. Original signals are modulated onto carrier frequencies, whose range must be acceptable by the link

2. Such a bandwidth range = 1 channel. Channels are separated by strips of unused bandwidth (guard bands) to prevent signal overlapping

<u>Combination step</u>: modulated signals are combined to produce a single composite signal that can be transported by the link

FDM: Demultiplexer



<u>Filtering step</u>:

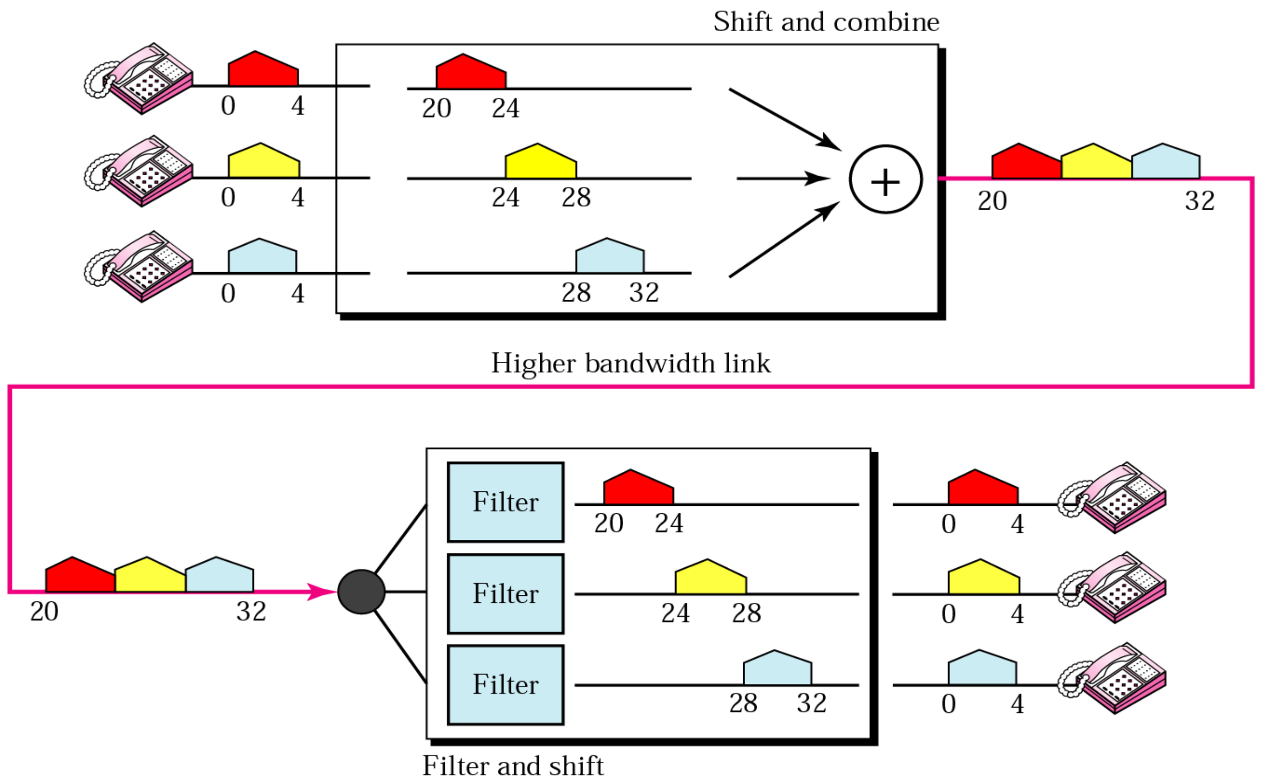
1. Decompose the composite signal into constituent component signals

<u>Demodulation step</u>:

1. Reconstruct the original signal from each modulated signal and send it to the corresponding receiver

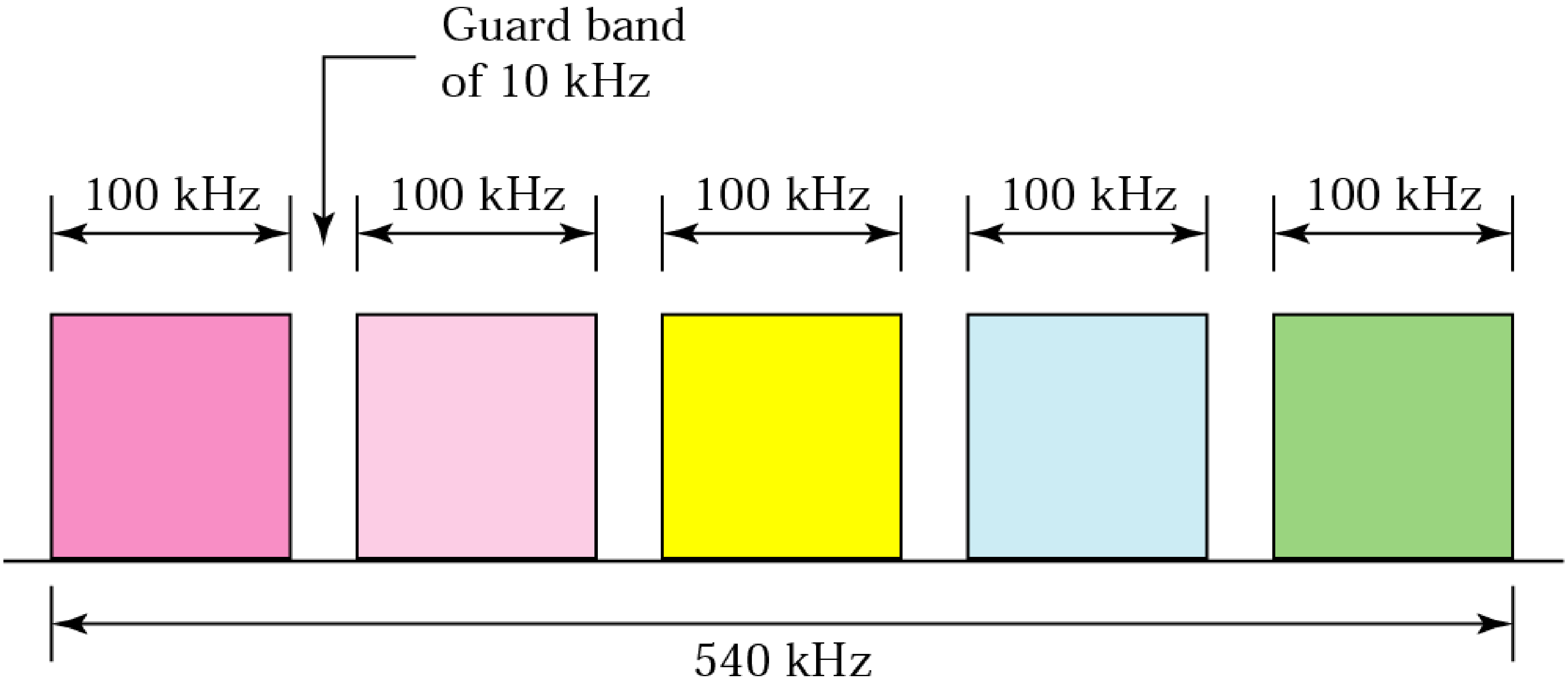
FDM: Example 1

Assume that a voice channel occupies a bandwidth of 4 KHz. We need to combine three voice channels into a link with a bandwidth of 12 KHz, from 20 to 32 KHz. Show the configuration using the frequency domain without the use of guard bands.

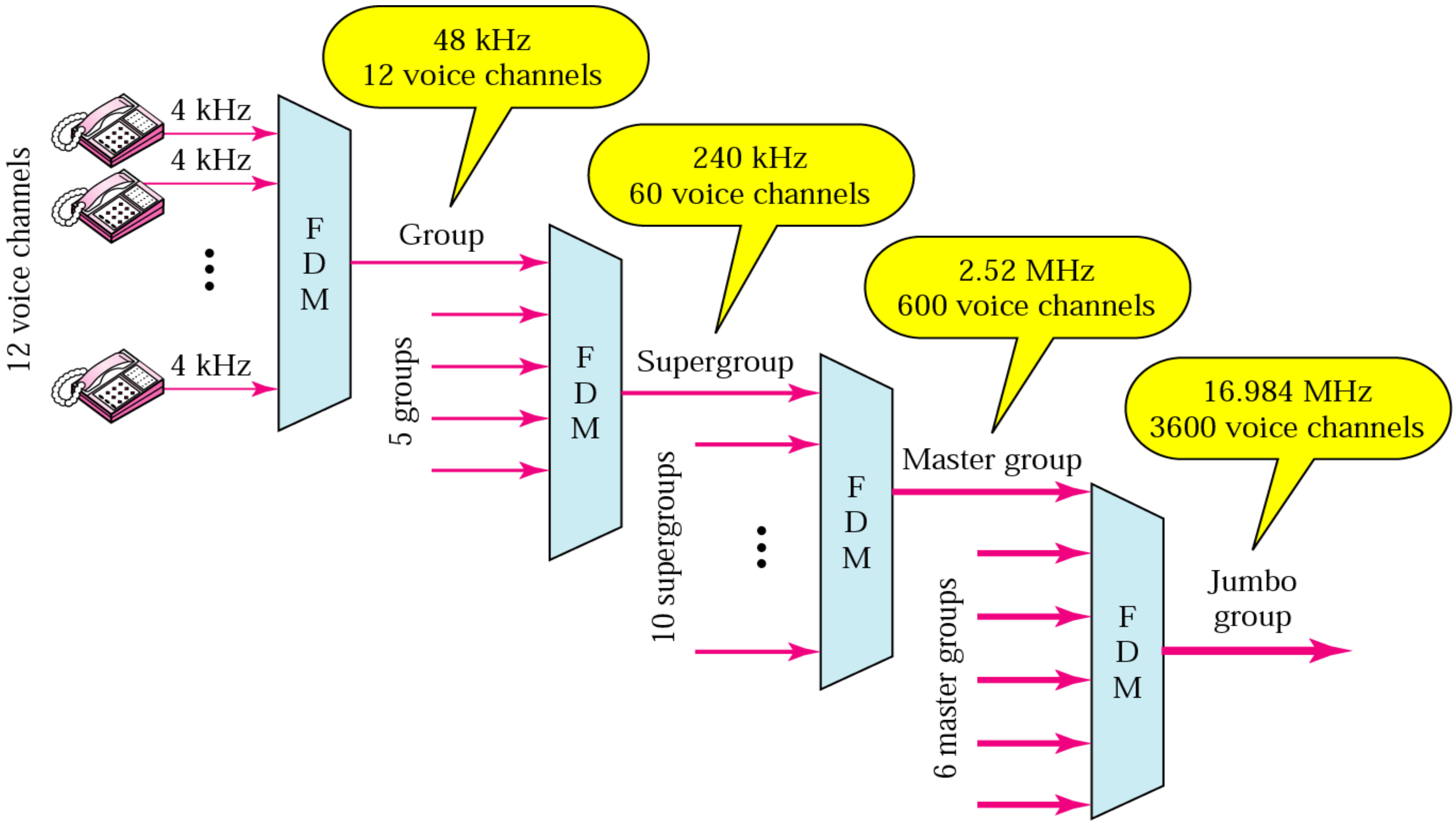


FDM: Example 2

Five channels, each with a 100-KHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 KHz between the channels to prevent interference?



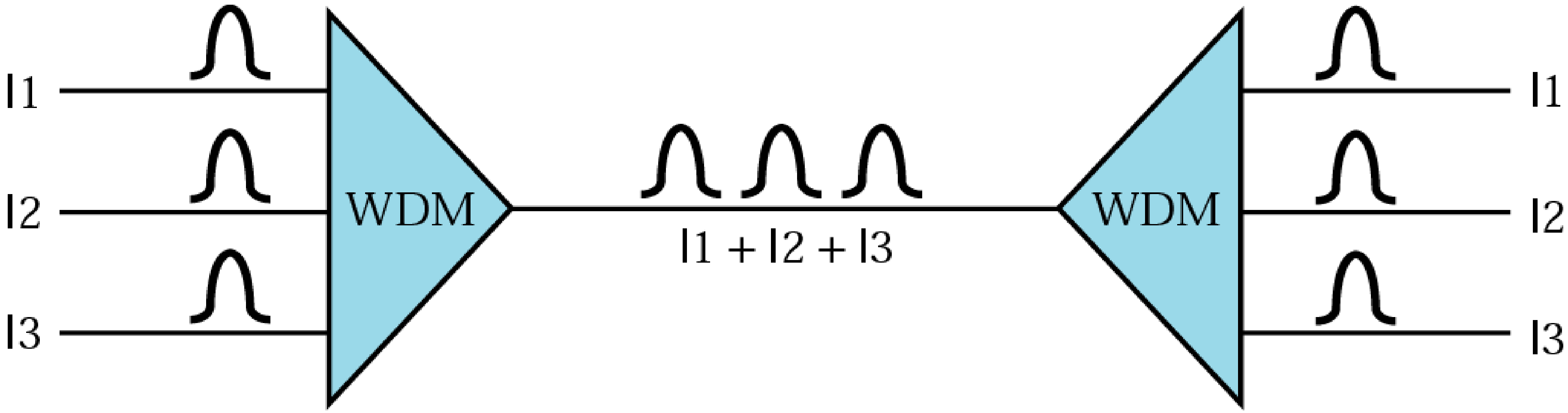
AT&T Analog Hierarchy
American Telephone & Telegraph



Wavelength-division multiplexing (WDM)

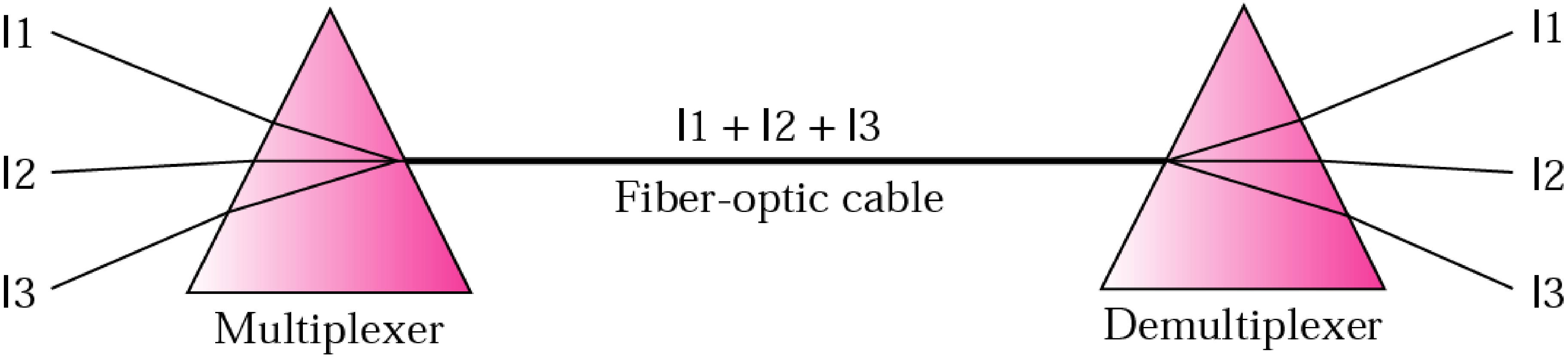
Wavelength-division multiplexing (**WDM**) is a technology which multiplexes a number of optical carrier signals onto a single optical fiber by using different wavelengths (i.e., colors) of laser light. This technique enables bidirectional communications over one strand of fiber, as well as multiplication of capacity.

Wave Division Multiplexing (WDM)



- **Optic fiber provides very high data rate (50,000Gbps in theory)**
- ***WDM is an analog multiplexing technique to combine optical signals to utilize the optical fiber***
- **A variation of FDM for optic channels**

WDM

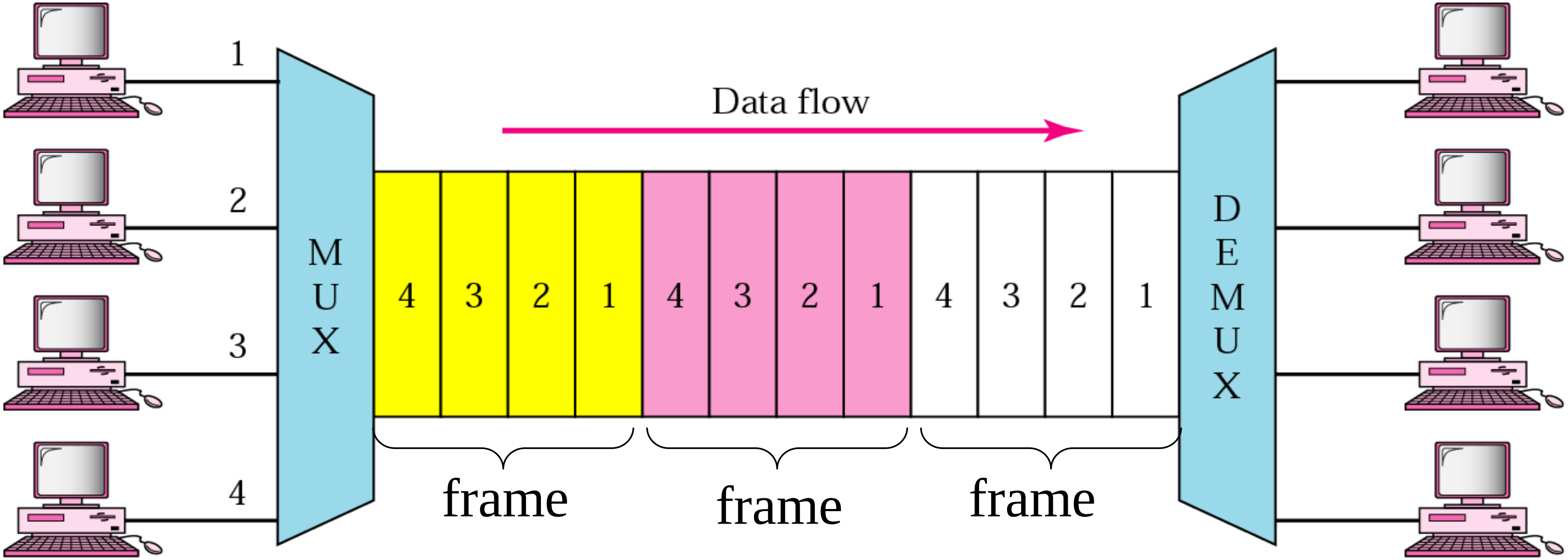


- Use prism to combine input light beams into one output beam of a wider band of frequency
  - Basic physics: prism bends a light beam based on the angle of incidence and the frequency
- Each output fiber contains a short, specially-constructed core that filters out all but one wavelength

Time-Division Multiplexing

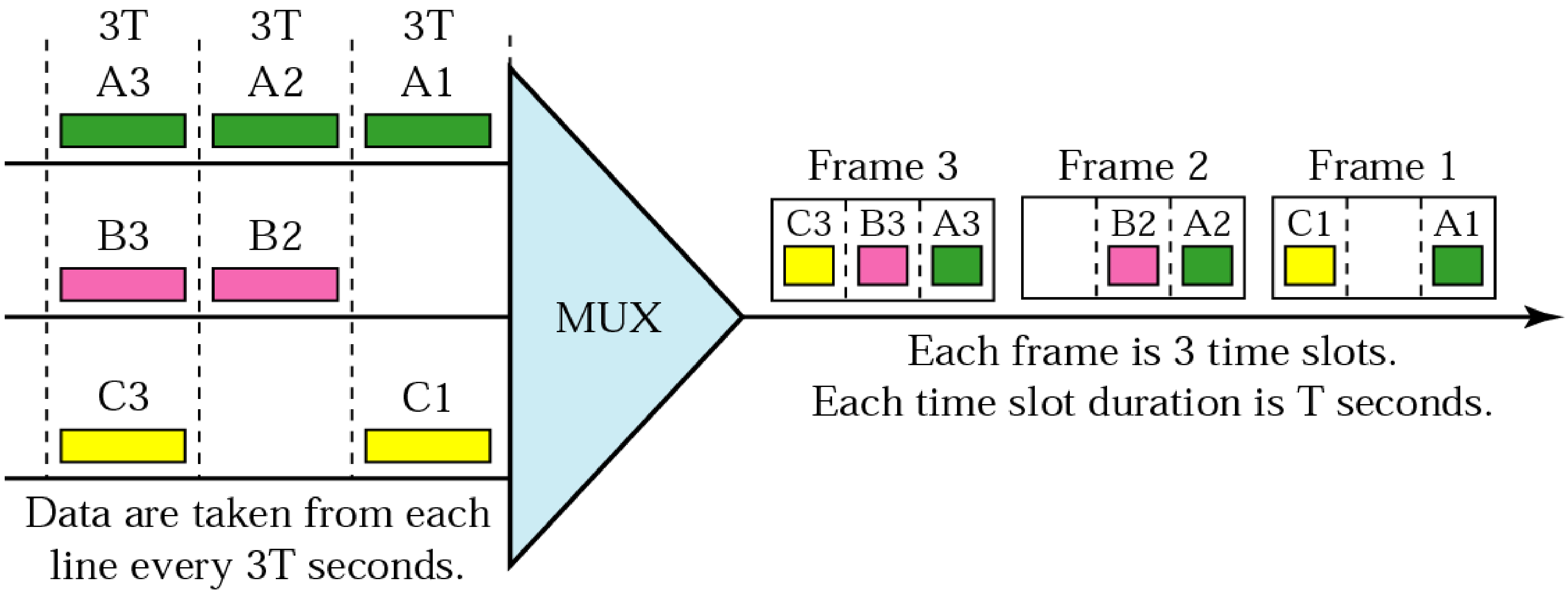
- Time division multiplexing (TDM) is a communications process that transmits two or more streaming digital signals over a common channel.
- In TDM, incoming signals are divided into equal fixed-length time slots.
- After multiplexing, these signals are transmitted over a shared medium and reassembled into their original format after de-multiplexing.
- Time slot selection is directly proportional to overall system efficiency.
- Time division multiplexing (TDM) is also known as a digital circuit switched.

Time-Division Multiplexing



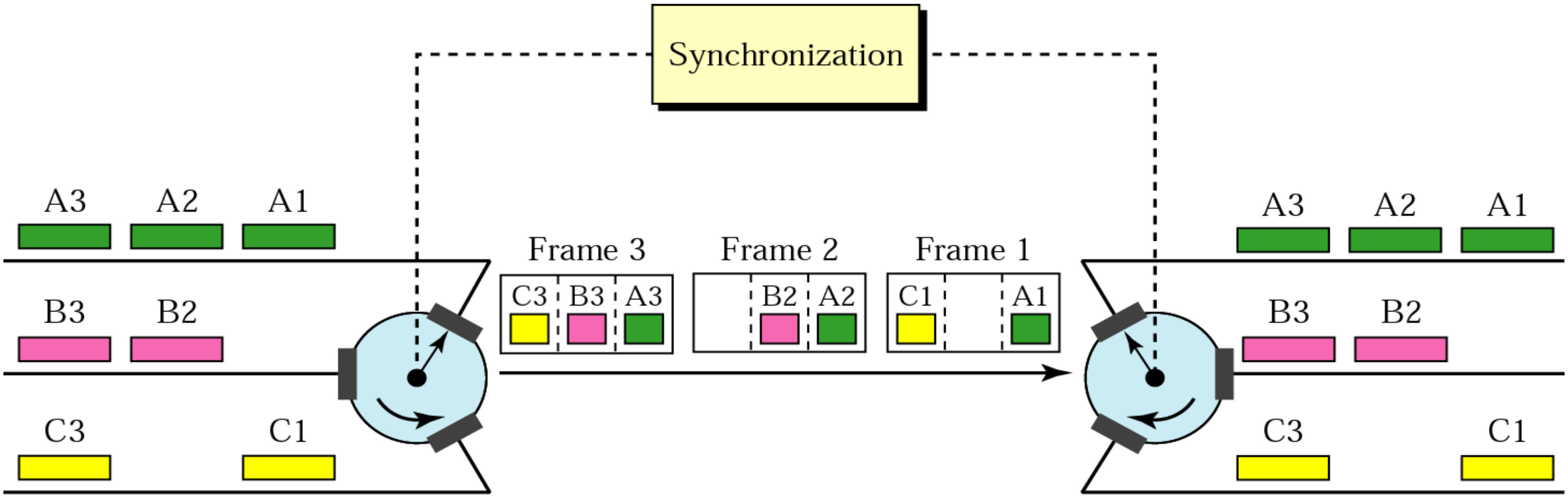
- TDM is a digital multiplexing technique to combine data.

TDM: Time Slots and Frames



- In a TDM, the data rate of the link is n times faster, and the unit duration is n times shorter.

TDM: Interleaving

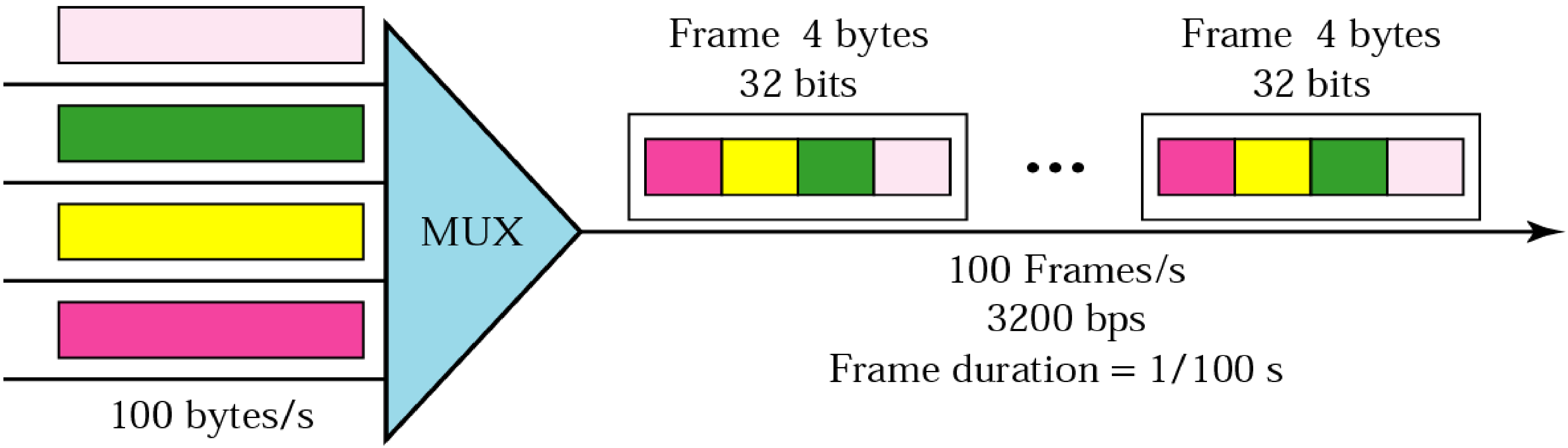


Mux: each connection in turn puts a data input into the path

Demux: each connection in turn receives a data unit from the path

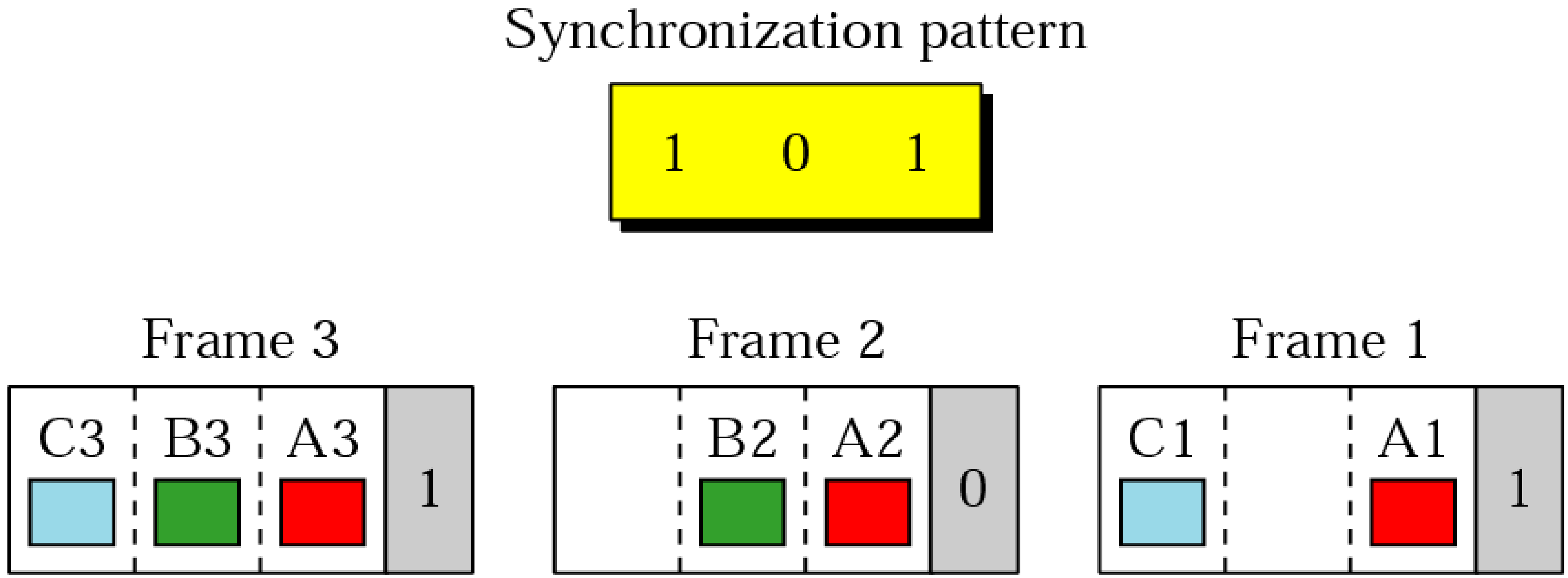
TDM: Example

4 channels are multiplexed using TDM. If each channel sends 100 bytes/s and we multiplex 1 byte per channel, show the frame traveling on the link, the size of the frame, the duration of a frame, the frame rate, and the bit rate for the link.



TDM: Framing Bits

- Mux and Demux may be out of sync => bits are delivered to wrong receivers
- Need to separate between frames => extra bits are inserted into the head of each frame. A bit pattern must be followed so that time slots are separated accurately.



TDM vs FDM

- Key Differences Between TDM and FDM
- The time-division multiplexing (TDM) includes sharing of the time through utilizing time slots for the signals. On the other hand, frequency-division multiplexing (FDM) involves the distribution of the frequencies, where the channel is divided into various bandwidth ranges (channels).
- Analog signal or Digital signal any could be utilized for the TDM while FDM works with Analog signals only.
- **Framing bits** (Sync Pulses) are used in TDM at the start of a frame in order to enable synchronization. As against, FDM uses **Guard bands** to separate the signals and prevent its overlapping.

TDM vs FDM

- FDM system generates different carriers for the different channels, and also each occupies a distinct frequency band. In addition, different bandpass filters are required. Conversely, the TDM system requires identical circuits. As a result, the circuitry needed in FDM is more complex than needed in TDM.
- The **non–linear** character of the various amplifier in the FDM system produces **harmonic distortion**, and this introduces the **interference**. In contrast, in TDM system time slots are allotted to various signals; as the multiple signals are not inserted simultaneously in a link.
- The utilization of physical link in case of TDM is more efficient than in FDM. The reason behind this is that the FDM system divides the link in multiple channels which does not make use of full channel capacity.

Thank You All.